01RUS9D2P4

SomeonecoolLovesMaths

May 31, 2026

Problem

Find all odd positive integers $n > 1$ such that if a and b are relatively prime divisors of n, then $a + b - 1$ divides n.

Solution. We claim that $n = p^k$ for some odd prime p are the only numbers that work. It is easy to check that they do work.

Now assume that there are at least two distinct primes dividing n. So, $n = p^k m$ such that $\gcd(p, m) = 1$ where p is the smallest prime dividing n.
We have $p + m - 1 \mid n$. If there is $q \neq p$ such that $q \mid p + m - 1$ then we will have that $q \mid p - 1$ since $q \mid m$. As a result, we have a prime smaller than p which divides n, contradicting the minimality. As a result, $p + m - 1 = p^a$ for some a.
So, $m - 1 = p^a - p$. Since $p^a + m - 1 = 2p^a - p$, we have $2p^{a-1} - 1 \mid m = p^a - p + 1$. This implies that $2p^{a-1} - 1 \mid p - 2$. Since p is a prime, we must have $a = 1$, or, $m - 1 = p - p = 0 \implies m = 1$. This of course contradicts the fact that n has at least 2 distinct prime divisors.